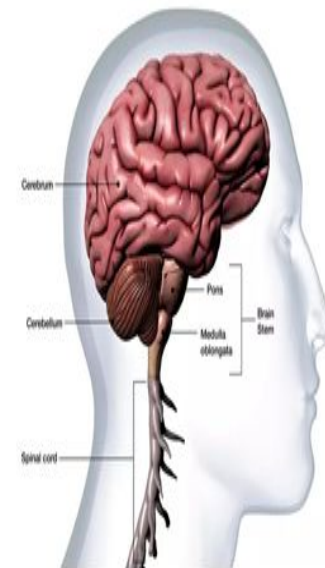
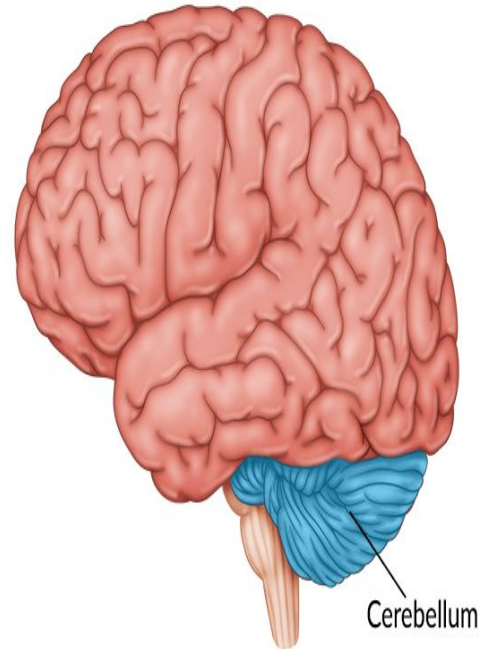
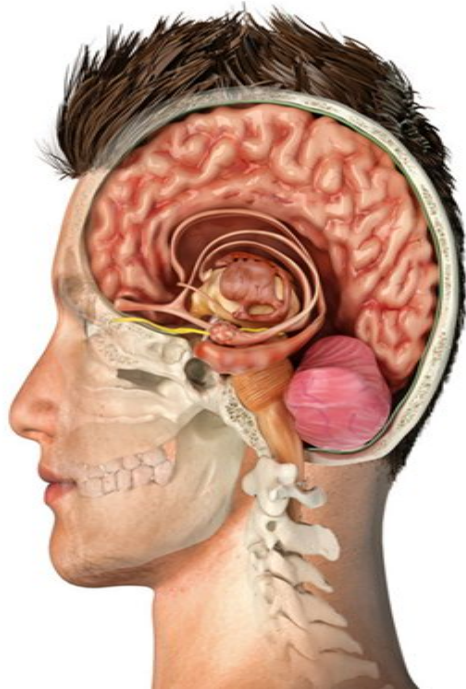


Cerebellum Functions



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CEREBELLUM

The cerebellum can be found just below **cerebrum** and **behind the upper portion of brain stem**.

This is the area at the base of skull where **head meets neck**.

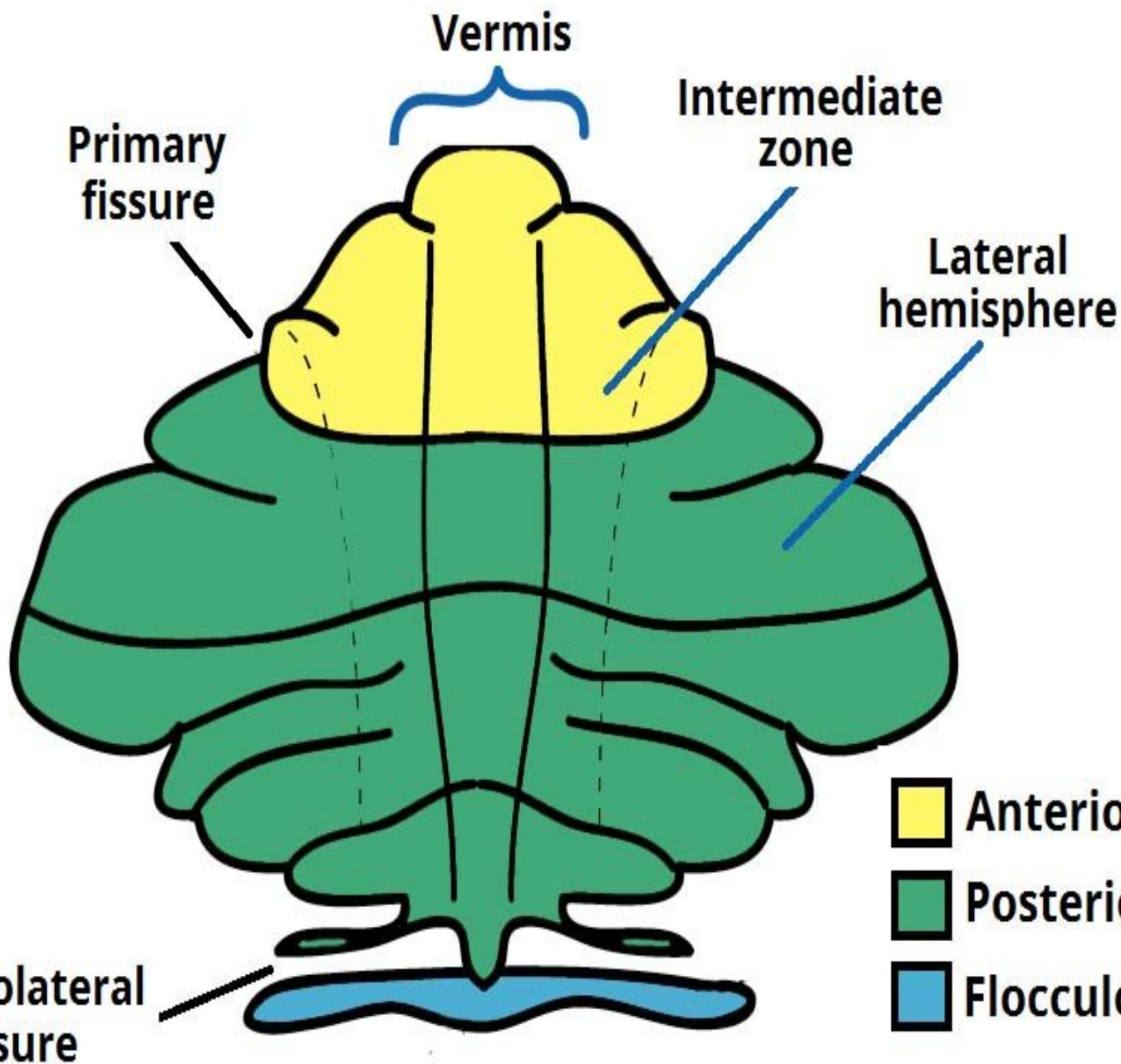
The cerebellum is divided up into **three different parts called lobes**.

These lobes are separated from each other by deep grooves called **fissures**.

There are two major components of the cerebellum:

Cerebellar cortex: This is a layer of thin, heavily folded tissue that contains most of the nerve cells in the cerebellum.

Cerebellar nuclei: Found deep within the cerebellum, the nerve cells of the cerebellar nuclei are primarily involved in sending information from the cerebellum.



Functions of the Cerebellum

The cerebellum controls voluntary movements such as:

- Walking
- Posture (Sense of body position)
- Balance and Equilibrium
- Fine movement Coordination
- Eye movements
- Speech

Cerebellum damage can be happen due to

Disruption of the cerebellum or its connections to other parts of the nervous system can happen in a variety of ways. For instance, the cerebellum can sustain damage due to:

- Head injury
- Stroke
- Brain tumor
- Autoimmune conditions, such as multiple sclerosis
- Neurodegenerative conditions, such as Parkinson's disease or Huntington's disease
- Infections
- Some medications, such as benzodiazepines or barbiturates
- Alcohol use disorder
- Heavy metal poisoning, such as due to lead or mercury

Cerebellum damage leads to following conditions

When the cerebellum is damaged, movement and balance can be affected. Damage to the cerebellum can result in conditions such as:

- **Ataxia:** loss of control of voluntary movement
- **Dystonia:** It is a movement disorder that causes the muscles to contract involuntarily. Patients with dystonia may experience uncontrollable twisting, repetitive movements or abnormal postures and positions
- **Tremors:** involuntary rhythmic contractions These are involuntary muscle contractions that occur in a rhythmic manner.
- **Vertigo:** It is the sensation of spinning. You may feel as if you're spinning or that your surroundings are spinning.

- **Dyslexia:**

It is a learning disorder in which a person has difficulty reading, spelling, or writing due to trouble processing how speech sounds relate to words or parts of words.

- **Anxiety disorders:**

It comprise a group of emotional disorders involving excessive levels of anxiety or fear.

- **Schizophrenia:**

It is a mental illness that has a variety of symptoms, such as hallucinations or delusions, a lack of emotion, and disorganized speech and movement.

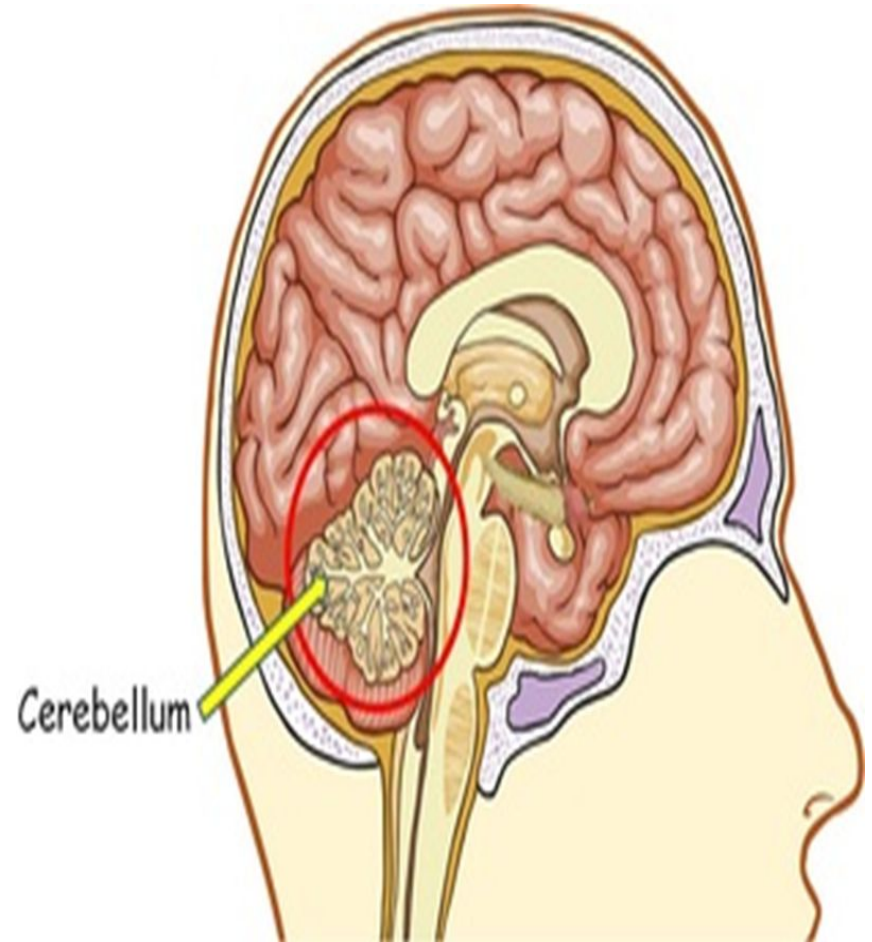
The function of cerebellar hemisphere can be assessed by

UPPER LIMB

- Finger nose test
- Finger to finger test
- Dysdiadochokinesia
- Rebounding
- Make a circle in the air
- Tone of the muscle
- Test for Tremors

LOWER LIMB

- Tandem walking
- Knee jerk
- The heel shin test
- Examination of gait
- **Romberg sign**
- **Test for Nystagmus**
- **Examination of Speech**



FINGER TO NOSE TEST

□ Patients are unable to touch the tip of their nose with their index finger with eyes closed

1. Ask the patient to alternatively point from his or her nose to the examiner's finger
2. The examiner will move his or her finger to different locations
3. Have the patient do this as rapidly as possible.
4. Also observe the **intention tremor**.



FINGER TO FINGER TEST

- Touch the tips of two index fingers with extended arms bringing them from a distance rapidly.



DYSDIADOCHOKINESIA

- **Dysdiadochokinesia**

It is defined as the inability to perform rapid alternating muscle movements.

- This means the inability to perform rapid alternating movements; e.g. pronation and supination of the hand.



REBOUNDING:

This is manifested by an overshooting of a limb when a resistance to its movement is suddenly withdrawn

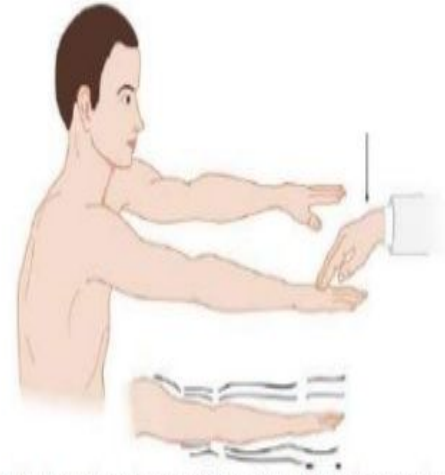


FIGURE 8-8. Wrist-tapping test for abnormal overshooting oscillation after sudden displacement of a part that is maintaining a volitional posture. The thin arrow shows the direction of the examiner's push, which displaces the part.



FIGURE 8-9. The arm-pulling test for overshooting. It tests how well the cerebellum functions to check movement and to maintain a given posture after a sudden release of tension on a muscle that is voluntarily contracting.

TREMORS

These are rhythmic movements resulting from alternating contracting and relaxation of groups of muscles.

These involve peripheral part of the body like hands, feet and tongue.

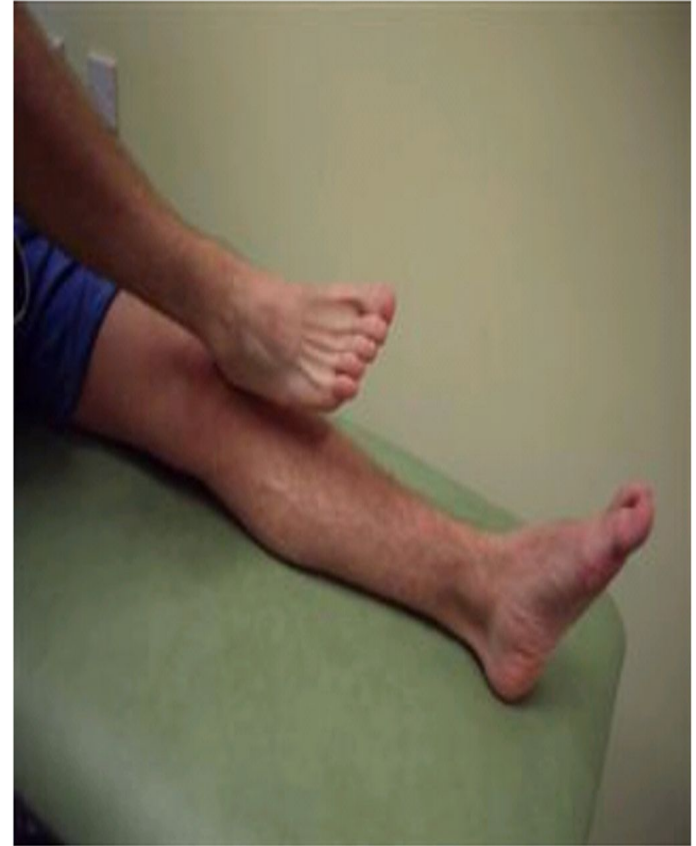
In order to look for tremors in the hand, ask the patient to outstretch the arms and abduct the finger.

If tremor are still and not obvious but suspected than place a piece of paper on the dorsum of hand and observe its movement



HEEL SHIN TEST

- Patient should lie in the bed.
- Ask him to place the heel on his opposite knee and slide it downward along the shin to the ankle, then lift it, making a semicircle, place it against on the knee and repeat the movement.
- As said earlier, movements are clumsy and jerky if coordination is lost.



Heel-shin (HTS test)

Examination of gait

TANDEM WALKING

- Ask the patient to walk along a straight line so that the heel of one foot comes directly in contact with the toes of the other foot.
- The rear foot is then advanced so that the heel is placed in the format of the previously front foot.
- Patient with cerebellar dysfunction is unable to do so and tends to fall towards diseased side.



ROMBERG'S SIGN

The Romberg test is a test that measures your sense of balance.

- The Romberg sign is present when a patient is able to stand with feet together and eyes open, but sways or falls with eyes closed.
- Ask the patient to stand with his feet close together.
- A patient of sensory ataxia is steady when eyes are opened and becomes unsteady when eyes are closed.
- A patient of cerebellar ataxia(motor ataxia) is equally unsteady whether eyes are opened or closed.



The Romberg test. Have the patient stand still with heels and toes together. Ask the patient to close her eyes and balance herself. If the patient loses her balance, the test is positive

DRUNKEN GAIT

Walking incoordination, is often described as a “drunken gait,”

- Ask the patient to walk in a straight line and observe the gait
- Patient walks on a broad base in a reeling manner.



TEST FOR NYSTAGMUS

□ Nystagmus is a vision condition in which the eyes make repetitive, uncontrolled movements

Nystagmus is a condition where the eyes move rapidly and uncontrollably. They can move:

- side to side (horizontal nystagmus)
- up and down (vertical nystagmus)
- in a circle (rotary nystagmus)



TEST:

- Ask the patient in front of him and observe whether the eyes remain steady.
- Now put a pencil or even your index finger in front of the patient at the level of eyes at a distance of 1 ½ feet.
- Move the object to the extreme left, then to the extreme right, then upward and downward gradually, and keep it for some time.
- In a positive case, there will be rhythmical oscillatory movements of the eye balls.

Nystagmus

COMMON WARNING SIGNS



**Uncontrollable
eye
movement**



**Nighttime
vision
problems**



**Shaky or
blurry vision**



**Balance
problems**



**Light
sensitivity**



Dizziness

EXAMINATION OF SPEECH

- Failure of progression occurs in Talking because the individual muscle movements in the larynx, mouth and respiratory system is not orderly.
- Ask him to say **artillery**, he will speak slowly and deliberately as **ar-til-ler-y**

PENDULOUS KNEE JERK

- Elicit knee jerk
- Leg after extension will fall down in a pendulous fashion while the patient is sitting on the bed with legs are hanging down.



Relevant Questions

Q-1-What are the functions of cerebellum?

The cerebellum primarily coordinates movement, balance, posture, and muscle tone, enabling smooth and precise voluntary actions.

It also plays a key role in motor learning, helping the brain learn and improve physical skills through feedback and error correction

Q-2-Name the nuclei and their pathway in spinal cord taken by stimuli from upper and lower limb to the cerebellum

A neural pathway is a bundle of axons that connects two or more different neurons, facilitating communication between them. Tracts are neural pathways that are located in the brain and spinal cord (central nervous system). Each tract runs bilaterally; one on each side of the cerebral hemisphere or in a hemisection of the spinal cord.

Stimuli from the lower limbs travel via the nucleus dorsalis (Clarke's nucleus) and the ventral spinocerebellar tract (from C8-L3 segments) or the dorsal (posterior) spinocerebellar tract (below these segments) to the ipsilateral cerebellum. Upper limb stimuli use the cuneate fasciculus and the Cuneo cerebellar tract for the ipsilateral pathway, or the rostral spinocerebellar tract for a contralateral pathway that decussates twice.

Q-3-What is the purpose of cerebellar function test?

The purpose of a cerebellar function test is to identify dysfunction in the cerebellum, the part of the brain responsible for coordinating voluntary movements, balance, and muscle tone.

These tests evaluate a patient's gait, balance, and ability to perform smooth, coordinated movements like the finger-to-nose test, and can help diagnose neurological conditions such as stroke, multiple sclerosis, or genetic disorders

Q-4-Name various tests used to check cerebellar functions of upper limb

Tests for upper limb cerebellar function include the Finger-to-Nose Test for coordinated movement and to detect intention tremor, and Dysdiadochokinesia (DDK) or Rapid Alternating Movements (RAM) tests to assess the ability to perform quick, alternating movements by tapping the hand back and forth. Additionally, the Rebound Phenomenon test checks for the ability to control muscle over-contraction, and a Draw-a-Circle test assesses the quality of fine motor control and smooth movement.

Q-5-Name various tests used to check cerebellar functions of lower limb

Tests for cerebellar lower limb function include assessing gait for balance and coordination, performing the heel-to-shin test for accurate limb movement, evaluating for dysdiadochokinesia by rapidly alternating movements, and observing Romberg's test to check balance and proprioception

Q-6-Does Romberg's test detect cerebellar function

No, Romberg's test does not directly detect cerebellar function; instead, it assesses proprioception and the vestibular system, distinguishing sensory ataxia (due to posterior column or peripheral neuropathy) from cerebellar ataxia.

A positive result, where the patient sways or falls only with eyes closed, indicates a deficit in proprioception or vestibular function, not the cerebellum.

Patients with severe cerebellar ataxia will be unsteady even with their eyes open, making the Romberg test unable to be performed

Q-7-What is meant by ataxia,dysmetria?what are three types of ataxia?

Dysmetria is a neurological symptom, often caused by cerebellar damage, characterized by an impaired ability to control the distance, speed, and range of motion of movements, leading to movements that overshoot (hypermetria) or undershoot (hypometria) their intended targets.

Types of Ataxia?

There are three main types of ataxia, which happen in different ways.

1-Cerebella,r 2-Sensory ,3-Vestibular:

Q-8-Are cerebellar lesions ipsilateral or contralateral?

Unlike the cerebral cortex, the cerebellum receives input from, and controls output to, the ipsilateral side of the body, and damage to the cerebellum therefore results in deficits to the ipsilateral side of the body.

Q-9-What is difference between medial and lateral cerebellar lesions?

Medial (vermis) cerebellar lesions primarily cause issues with balance and gait (trunk and leg control), while lateral (hemispheric) lesions primarily cause limb incoordination (arm and hand movements), including intention tremor and past pointing.

Autism spectrum disorder: ASD is a developmental condition characterized by impairments in communication and social interaction as well as repetitive or restricted behaviors.



i DOES MY CHILD HAVE AUTISM (ASD)?

Autism (ASD) may present with a variety of symptoms, but the core affected areas are:

Deficits in language and communication

Impairments in socialization and social interactions

Undeveloped cognitive and adaptive functioning

Restricted repetitive and stereotyped patterns of behavior and interests

